

Highlights

Threshold Selection as a Critical Design Choice in Financial Fraud Detection: A Systematic Comparison of Classical Machine Learning Models and Tabular Transformers

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- Validation-only threshold selection separates ranking from alert decisions.
- Higher fraud recall can require a substantially larger alert workload.
- Seven imbalance strategies show model- and dataset-dependent effects.
- Boosting and a tabular transformer are compared under explicit protocol limits.
- Saved results support an auditable, evidence-bounded decision framework.

Threshold Selection as a Critical Design Choice in Financial Fraud Detection: A Systematic Comparison of Classical Machine Learning Models and Tabular Transformers

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ABSTRACT

Fraud detectors must both rank suspicious cases and convert scores into operational alerts. These tasks are often conflated when model comparisons rely on a fixed decision threshold. We examine their relationship through an evaluation with development-only model and threshold selection, comparing Logistic Regression, Random Forest, LightGBM, CatBoost, FT-Transformer, and a One-Class SVM reference on ULB credit card transactions and BAF Base bank account applications. A seven-strategy imbalance study covers the five supervised models; a separate four-rule threshold study holds each baseline model's test scores fixed. All model and threshold selection is confined to development data, using five-fold validation for classical models and an internal holdout for FT-Transformer. Exact predictor-profile deduplication precedes the ULB split. LightGBM attains the highest baseline ULB PR-AUC (0.8205); CatBoost leads BAF at 0.1796, followed by FT-Transformer (0.1769) and LightGBM (0.1768). Nevertheless, the fixed threshold of 0.5 yields BAF F_2 scores of only 0.001–0.048 for supervised models, compared with 0.287–0.316 under validation-selected max- F_2 thresholds. For FT-Transformer, this is approximately a 6.3-fold increase, accompanied by an alert-rate increase from 0.07% to 5.71%. No imbalance intervention consistently improves ranking performance; SMOTE reduces BAF FT-Transformer PR-AUC from 0.1769 to 0.1149, with preprocessing differences preventing an architecture-only interpretation. The contribution is an auditable separation of ranking, threshold selection, and review workload, together with evidence-bounded configuration guidance. These benchmark results support explicit validation-based operating-point selection, not a universal classifier or resampling prescription.

1. Introduction

Machine learning is an established component of financial fraud detection, but useful ranking performance does not by itself specify which cases should be investigated (Ryman-Tubb et al., 2018). A deployed system must translate continuous scores into alerts, balancing missed fraud against false alerts, investigator capacity, and customer friction. Under severe class imbalance, this distinction is important: a model may rank fraud above legitimate activity while assigning relatively few cases a score greater than the conventional threshold of 0.5.

Reliable evaluation requires separating three decisions: how the model is fitted, how its operating threshold is selected, and how the resulting detector is assessed. Resampling before the train/test split can contaminate evaluation (Hayat and Magnier, 2025). Selecting a threshold on test labels introduces another selection bias. In contrast, retaining the fixed threshold of 0.5 is not inherently leakage, but may produce an unsuitable operating point. These are different problems and should not be treated as interchangeable explanations for poor or optimistic results. Similarly, ROC-AUC describes discrimination over false-positive rates, whereas precision-recall analysis exposes the positive predictive value at the observed prevalence (Saito and Rehmsmeier, 2015).

This paper asks how much the operational conclusions of a fraud benchmark change when threshold selection is made explicit and kept outside the test set. We compare five supervised model families and a One-Class SVM (OCSVM) reference on two datasets with different feature representations, sample sizes, and fraud processes. We then separate a seven-strategy imbalance experiment from a four-rule threshold experiment using fixed baseline test scores. This design

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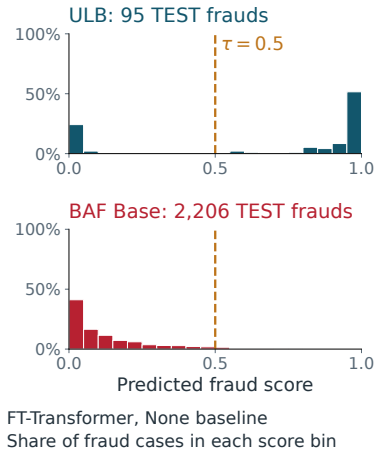
distinguishes changes in learned rankings from changes in the decisions made from those rankings; it does not assume that a higher ranking score automatically yields a better operational detector.

The contribution is empirical and methodological, rather than a new classifier or threshold optimiser. First, the study provides a common outer holdout and explicit model-specific inner validation for classical and transformer-based models. Second, it quantifies threshold sensitivity jointly with alert workload, so that gains in F_2 are not presented without their investigation burden. Third, it evaluates all seven imbalance strategies for the five supervised models on both datasets while reusing baseline-selected hyperparameters. Finally, it translates only evaluated configurations into conditional practitioner guidance. No universal superiority claim is inferred from these two benchmarks.

Figure 1 introduces the experimental logic and its practical consequence. Its numerical example uses the saved BAF FT-Transformer threshold study: changing the operating rule changes alerts and F_2 without changing the test-score ranking. The figure also makes explicit that the classical and transformer validation paths differ, a distinction carried through the analysis.

From fraud scores to validated alert decisions

1 Inspect saved fraud scores



2 Select within DEV only

80% DEV / 20% TEST

ULB: deduplicate before split
Stratified holdout, seed 42

Model-specific validation

Classical / OCSVM: five folds
FT-Transformer: internal holdout

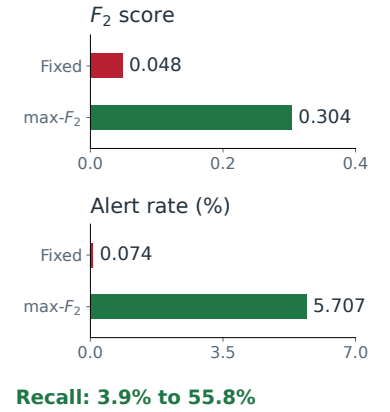
Freeze the operating rules

Reuse saved validation scores
Four rules, unchanged TEST ranking

TEST excluded from selection

3 Quantify the trade-off

BAF Base / FT-Transformer



Validate the operating point and its workload; do not infer decisions from ranking alone.

Figure 1: From score distributions to operational decisions. Left: empirical histograms of saved FT-Transformer scores for fraudulent TEST cases, normalised within each dataset; these are not schematic distributions or a claim about all models. Centre: development-only selection with a protected outer TEST partition. Right: BAF FT-Transformer results for two rules applied to the same saved TEST scores. The max- F_2 rule raises recall and alert workload together. Layout and plotting code were revised with OpenAI Codex assistance. Numerical panels are derived reproducibly from saved evaluation artefacts, as described in Section 3.7.

Section 2 positions the study in the literature. Section 3 specifies the protocol and its boundaries. Section 4 reports the numerical evidence; Section 5 interprets it and develops configuration guidance. Section 6 concludes.

2. Related Work

2.1. Fraud Benchmarks and Evaluation

ULB credit card data remain a widely used reference for fraud detection (Dal Pozzolo et al., 2015, 2018). Its 28 PCA-transformed predictors protect confidentiality but limit domain interpretation; Time and Amount retain their original meanings. The Bank Account Fraud (BAF) suite instead provides synthetic account-opening applications with semantic numerical and categorical features, together with variants for studying controlled distributional differences (Jesus et al., 2022b). We use BAF Base only. Comparing ULB with BAF therefore changes more than sample size: the fraud task, feature representation, prevalence, and data generation process also differ.

These benchmarks require metrics appropriate to rare positives. Saito and Rehmsmeier (2015) explain why precision–recall analysis can reveal behaviour not apparent from ROC curves in imbalanced settings. ROC-AUC is not intrinsically invalid, but does not by itself express the positive predictive value or workload at a deployment threshold. We retain it as a secondary discrimination metric and use average precision, reported as PR-AUC, for ranking. Threshold-dependent F_1 , F_2 , recall, and alert rate describe operating decisions. None of these replaces a domain-specific monetary cost model, which is outside the present evidence.

2.2. Imbalance Interventions and Tabular Models

The imbalance literature distinguishes data-level interventions from loss-level weighting (He and Garcia, 2009). SMOTE interpolates minority examples (Chawla et al., 2002); its combinations with Tomek Links and Edited Nearest Neighbours add cleaning stages (Batista et al., 2004). Their effectiveness depends on the data representation and learner, and synthetic interpolation can be problematic in heterogeneous feature spaces (Fernández et al., 2018). Consequently, an unmodified baseline is necessary: an intervention is not beneficial merely because it makes training classes balanced.

For tabular learning, gradient boosting provides a strong comparator to neural architectures. Shwartz-Ziv and Armon (2022) report competitive performance of boosted trees against the deep tabular models they study. Gorishniy et al. (2021) introduce FT-Transformer and evaluate a systematic neural baseline. These studies motivate including both boosting and a transformer; they do not imply a predetermined winner in fraud detection. In particular, a comparison of architectures is also a comparison of implemented preprocessing and validation choices unless these are explicitly controlled.

2.3. Threshold Selection as a Separate Decision

Work on fraud decision-making distinguishes classifier outputs from subsequent business actions (Luzio et al., 2024). Probability estimation and operating-point selection are related but not equivalent: a threshold can be selected from validation scores without applying probability calibration, and the score of 0.5 need not maximise a recall-weighted criterion. The meaning of a threshold also depends on the output scale; an OCSVM anomaly score is not a fraud probability.

Our study builds on this separation by reporting the same baseline test scores under four pre-specified threshold rules, alongside a separate imbalance factorial. The value of this design is the traceability of each comparison. Within the threshold study, the ranking is fixed and only the cut changes. Within the imbalance study, the intervention changes while baseline-selected hyperparameters are retained. This supports a bounded account of which observed gains arise at the decision stage and which require a different fitted pipeline. It is not a claim to have invented validation-based threshold selection.

3. Experimental Framework

3.1. Datasets

Table 1 summarises the two datasets. The raw ULB file contains 284,807 European credit card transactions from two days in September 2013, including 492 frauds (0.1727%). Its predictors comprise V1–V28, Time, and Amount. Before splitting, exact duplicates across all 30 predictors are removed, retaining the first original occurrence. Conflicting labels for an identical profile would cause the check to fail. The supplied file contains no such conflict: 1,081 repeated rows, including 19 frauds, are removed, leaving 283,726 distinct profiles and 473 frauds. Original row indices are retained for auditing, never as predictors. Thus ULB evaluation concerns distinct observed profiles rather than their raw transaction-frequency distribution. BAF Base contains 1,000,000 synthetic bank account applications, including 11,029 frauds (1.1029%). The implemented BAF input excludes the month field and uses 30 predictors: 25 numerical and five categorical (payment_type, employment_status, housing_status, source, and device_os). BAF records are applications, not payment transactions; cases denote either unit when discussing both datasets.

Table 1
Modelling populations after the ULB exact-profile integrity check.

Dataset	N	Fraud (%)	Features	Type	Reference
ULB 2013	283,726	0.17	30	PCA + Time/Amount	Dal Pozzolo et al. (2018)
BAF Base	1,000,000	1.10	30	Semantic	Jesus et al. (2022b)

3.2. Models and Preprocessing

The supervised comparison includes Logistic Regression (LR), Random Forest (RF) (Breiman, 2001), LightGBM (LGBM) (Ke et al., 2017), CatBoost (Prokhorenkova et al., 2018), and FT-Transformer (Gorishniy et al., 2021). OCSVM (Schölkopf et al., 2001) provides a separate anomaly-detection reference. LR uses the `saga` solver. In the revised ULB environment (scikit-learn 1.8), `C` and `l1_ratio` control the regularisation strength and Elastic Net mixture. The preserved BAF fit is reported with its saved configuration rather than interpreted as a newly retuned regularisation experiment.

Classical models and OCSVM share a preprocessor that mean-imputes and standardises numerical variables and mode-imputes and one-hot encodes categorical variables. CatBoost’s native categorical handling is not used. FT-Transformer mean-imputes and standardises numerical inputs and uses ordinal category codes as integer embedding indices. Preprocessors are fitted only on the corresponding training partition. OCSVM is fitted on legitimate training cases only; labelled validation data are nevertheless used to select its threshold. It is therefore a one-class fitted reference, not a wholly label-free workflow.

The BAF datasheet identifies `-1` absence codes in the previous- and current-address duration, bank-account duration, session duration, and device distinct-email count, and negative values of `intended_balcon_amount` as absent information (Jesus et al., 2022a). The primary representation preserves these numerical codes; mean imputation affects actual NaN values only, and standardisation includes the codes in training-partition statistics. Trees can split on absence codes, whereas linear and neural models receive coded numerical magnitudes. Other negative measurements are not indiscriminately recoded. A predefined BAF sensitivity analysis changes only the documented codes to NaN, imputes training means and adds missingness indicators. It retains the same split and selected hyperparameters and selects its thresholds within DEV; it does not choose a replacement primary policy using TEST performance.

3.3. Imbalance Strategies

The five supervised families are evaluated under None (unmodified baseline), Random Under-Sampling (RUS), Random Over-Sampling (ROS), SMOTE, SMOTE+Tomek Links (SM+T), SMOTEENN, and Class Weights. The samplers use their implemented default balancing settings, with random seed 42. Class weighting uses balanced class weights for classical models and the negative-to-positive training-count ratio in the transformer’s weighted binary cross-entropy. OCSVM is excluded from this factorial.

Resampling takes place after preprocessing and only within training partitions. For classical models, synthetic interpolation therefore operates on standardised numerical and one-hot categorical columns. For FT-Transformer, it operates on concatenated numerical values and ordinal category codes; synthetic categorical values are subsequently converted to integer embedding indices. Neither primary path uses a categorical-aware sampler. A separate predefined BAF control compares primary SMOTE with SMOTE-NC for CatBoost and FT-Transformer at fixed selected hyperparameters. CatBoost still receives one-hot encoded classifier inputs. Differences under SMOTE consequently cannot be attributed to architecture alone.

The primary design contains $7 \times 5 \times 2 = 70$ supervised model–strategy–dataset configurations plus two OCSVM references. Each uses its own development-selected max- F_2 threshold. The four-rule threshold comparison described next is restricted to the None baselines; it is not a $7 \times 5 \times 4 \times 2$ factorial.

3.4. Threshold Rules and Metrics

A case is flagged when its score $s(x) \geq \tau$. The primary ranking metric is `average_precision_score`, denoted PR-AUC throughout; it is average precision, not trapezoidal integration of the displayed PR curve. ROC-AUC is secondary. For precision P and recall R ,

$$F_\beta = \frac{(1 + \beta^2)PR}{\beta^2 P + R}. \quad (1)$$

We report both F_1 and recall-weighted F_2 . The choice $\beta = 2$ is a study criterion, not an estimate of a monetary false-negative cost. Alert rate is $(TP + FP)/N$ and expresses the proportion of cases requiring review.

The four pre-specified rules are fixed $\tau = 0.5$, validation max- F_1 , validation max- F_2 , and a validation precision target of at least 0.5. For the latter, the implementation selects the lowest observed validation threshold satisfying the target within each validation fold; if none qualifies, it sets $\tau = +\infty$ and issues no alerts. The target is not a guarantee of TEST or future precision, nor a fixed alert-budget constraint. For OCSVM we use the negated decision function, so higher scores indicate greater anomaly; its fixed 0.5 reference has no probabilistic interpretation.

3.5. Development-Only Selection and Final Evaluation

Each modelling population is partitioned once into stratified 80% DEV and 20% TEST sets using seed 42. Deduplicated ULB contains 226,980 DEV profiles (378 frauds) and 56,746 TEST profiles (95 frauds); BAF contains 800,000 DEV applications (8,823 frauds) and 200,000 TEST applications (2,206 frauds). Exact common-predictor overlap is zero across the outer splits, and across all ULB inner partitions. This verifies profile separation, not independence between unobserved customers or entities. BAF months are pooled rather than split chronologically, so the experiment is not a prospective temporal test. TEST is not used to fit preprocessing, resample training data, tune hyperparameters, select epochs, or select thresholds.

For classical models and OCSVM, stratified five-fold validation operates within DEV. Each fold fits its own preprocessing and model on its training subset, applying any resampling there alone. Thresholds are selected from the corresponding validation scores and aggregated by their median. The final pipeline is fitted on DEV, and the selected threshold is applied unchanged to TEST. For OCSVM, training is restricted to legitimate cases while validation retains both classes.

FT-Transformer instead uses one stratified 80/20 split inside DEV. This internal holdout supports tuning, early stopping, and threshold selection. After selection, the model is fitted on full DEV for the selected number of epochs. The outer TEST protection is shared with the classical path, but the inner validation estimators are not identical.

The separate threshold study uses the same saved validation scores and final TEST scores as the baselines. Classical fold thresholds are aggregated by their median, without re-optimisation on pooled validation or TEST. FT-Transformer’s four rules use the scores retained from its exact selected validation checkpoint; no additional validation model is reconstructed for this analysis. Ties under $\max-F_1$ or $\max-F_2$ are resolved towards the smallest qualifying threshold. The study’s $\max-F_2$ threshold and confusion counts must reproduce the baseline exactly; a mismatch causes reporting to fail. BAF classical validation scores were recovered with fixed hyperparameters while preserving the original final models and TEST score arrays. This recovery is documented separately from the threshold comparison and is not a new hyperparameter search.

3.6. Hyperparameter Optimisation

Each supervised baseline search has a budget of 50 attempted Optuna trials, including pruned trials, with a TPE sampler and a MedianPruner (Akiba et al., 2019), optimising PR-AUC on DEV validation. Classical models use five folds; FT-Transformer uses the internal holdout. During hyperparameter search only, LightGBM and CatBoost use early stopping with patience 50 rounds on their default binary-loss metric; average precision is the external search objective. The selected folds’ median best-iteration count fixes the tree count in subsequent validation and full-DEV fits. FT-Transformer uses patience 15 epochs on validation PR-AUC. All ULB supervised baselines are searched afresh after deduplication. BAF retains its previously selected baseline hyperparameters: the classical final fits are preserved and FT-Transformer is refitted without a new search. For the transformer, a seeded validation refit selects the best average-precision checkpoint, threshold and epoch count. A fresh seed-42 full-DEV model uses that epoch count, with the same 200-epoch AdamW/cosine-scheduling horizon in validation and final fitting. OCSVM uses a fixed configuration rather than a 50-trial search.

The selected baseline hyperparameters are reused across imbalance interventions within each model and dataset. This limits computational cost and isolates the implemented intervention from another hyperparameter search, but does not find the best possible separately retuned configuration for every strategy. Exact search spaces are defined in the repository’s model modules; each saved run’s `config.json` records its selected parameters and split provenance (Section 3.7).

3.7. Uncertainty, Computational Cost, and Reproducibility

Saved 95% percentile bootstrap intervals use 1,000 resamples of the fixed TEST predictions, skipping single-class resamples. These describe test-sample uncertainty conditional on the fitted model and split. They do not include variability from training seeds, different splits, or hyperparameter searches, and overlapping marginal intervals are not a paired equivalence or significance test.

Experiments ran on an Intel Core i5-13600KF (14 cores), 16 GB RAM, and an NVIDIA RTX 3070 (8 GB VRAM). Historical records used Python 3.12.7; the corrected October workflow uses Python 3.12.14. Classical models used CPU implementations; FT-Transformer used PyTorch on GPU. Reported times are recorded wall-clock tuning, final training, and full TEST inference times, not deployment latency guarantees. Historical BAF classical tuning and final-fit measurements are identified separately from current validation recovery and FT refitting. CPU/GPU execution,

inner-validation workloads and recorded search invocations differ; these are workflow measurements, not intrinsic architectural speed rankings.

The implementation and compact aggregate evidence are available at <https://github.com/01avo200100274/FraudDetection>. The repository documents environment setup and reproduction commands. In the authors' audit archive, corrected evidence is pinned by `results_revision/20261005/revision_manifest.json` and its derived `reporting/generation_qa.json`. These identify 72 primary configurations, 12 baseline threshold studies and the separate sensitivity controls. Some BAF classical artefacts are explicitly retained from `results/`; they are not selected by directory recency. Configuration and metric JSON files, saved labels and scores, PR curves, row identities and SHA-256 hashes document the evaluated fits. The compact public export instead provides aggregate metrics, tables, figures and portable provenance under `results/`; it does not distribute the saved per-row labels, scores, row-index arrays or fitted models. The article reuses the archived corrected evidence without fitting additional models or replacing experimental records. Additional archive-access or deposit arrangements require author confirmation and the applicable dataset terms.

The study-overview figure is generated by the accompanying `build_study_overview.py`. Its histograms use saved baseline TEST labels and scores with common, fixed-width bins on $[0, 1]$, normalised within fraudulent cases separately for each dataset. Its BAF operating-point comparison reads `threshold_study.json` directly. OpenAI Codex assisted with diagram layout and plotting code; no generative image model supplied the data panels. The companion `build_article_figures.py` reproduces the PR curves and imbalance heatmaps from run paths pinned in `figs/figure_sources.json`, using the same recorded values.

4. Results

4.1. Baseline Model Comparison

Tables 2 and 3 report the unmodified None baselines at their validation-selected max- F_2 thresholds. The ranking metrics do not depend on those thresholds. On deduplicated ULB, LGBM has the highest baseline PR-AUC point estimate (0.8205), closely followed by CatBoost (0.8196), FT-Transformer (0.8015), RF (0.7928), and LR (0.6921). LGBM also has the highest baseline F_2 (0.8134), whereas RF's F_2 (0.7883) exceeds CatBoost's (0.7751). Model preference therefore depends on whether the question concerns ranking or a particular operating criterion, even when the same fitted scores are used.

Table 2

Baseline performance on ULB 2013 (strategy = None, threshold = max- F_2).

Model	PR-AUC	ROC-AUC	F_1	F_2	Recall	Alert Rate
LR	0.692	0.957	0.791	0.771	0.758	0.15%
RF	0.793	0.929	0.820	0.788	0.768	0.15%
LGBM	0.820	0.980	0.852	0.813	0.789	0.14%
CatBoost	0.820	0.980	0.821	0.775	0.747	0.14%
FT-Trans.	0.801	0.967	0.800	0.761	0.737	0.14%
OCSVM	0.259	0.922	0.359	0.485	0.632	0.42%

Table 3

Baseline performance on BAF Base (strategy = None, threshold = max- F_2).

Model	PR-AUC	ROC-AUC	F_1	F_2	Recall	Alert Rate
LR	0.143	0.875	0.187	0.287	0.445	4.14%
RF	0.159	0.878	0.205	0.294	0.415	3.36%
LGBM	0.177	0.898	0.222	0.316	0.439	3.25%
CatBoost	0.180	0.898	0.211	0.314	0.468	3.80%
FT-Trans.	0.177	0.896	0.181	0.304	0.558	5.71%
OCSVM	0.019	0.628	0.045	0.081	0.171	7.32%

On BAF, CatBoost leads at PR-AUC 0.1796, followed by FT-Transformer (0.1769) and LGBM (0.1768). RF attains 0.1588 and LR 0.1432. These are close top-tier point estimates, not exact ties or evidence of statistical equivalence. OCSVM is substantially weaker on both datasets (0.2593 on ULB and 0.0190 on BAF), supporting its role as a reference rather than a competitive recommended configuration in this experiment.

Figure 2 displays baseline PR curves with the four threshold-study operating points. The important contrast is not that one dataset has universally separable scores and the other does not, but that the fixed 0.5 rule occupies very different recall regimes. On BAF it yields little recall for every supervised baseline. On ULB it is closer to useful operating points, although LR still benefits appreciably from lowering the threshold. A comparison between datasets cannot isolate a sample-size effect because their fraud tasks and representations also differ.

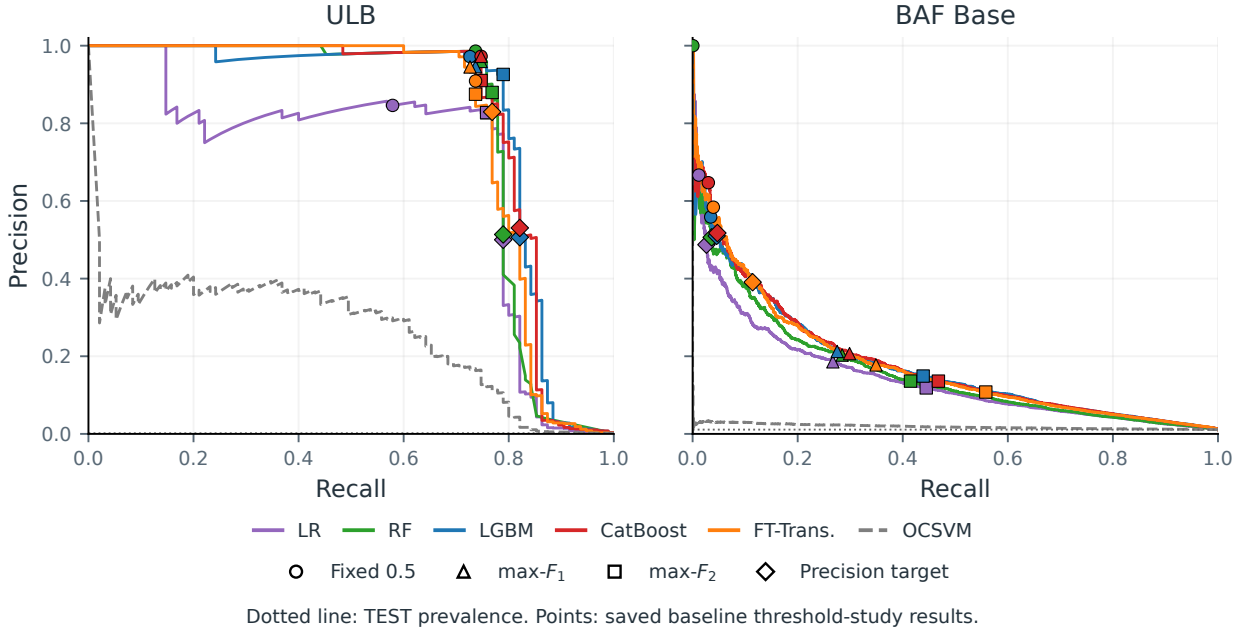


Figure 2: Baseline PR curves on ULB (left) and BAF Base (right), with threshold-study operating points: circle, fixed 0.5; triangle, max- F_1 ; square, max- F_2 ; diamond, validation Precision ≥ 0.5 . Markers use the same saved validation checkpoints and TEST scores as the baseline threshold study. OCSVM is a dashed ranking reference; its markers are omitted because its anomaly-score scale is not probabilistic. Plotting code was prepared with OpenAI Codex assistance and uses the saved experimental records.

4.2. Threshold Sensitivity and Alert Workload

Tables 4 and 5 compare rules on fixed baseline TEST scores. Table 6 reports their BAF alert workload. The rule names specify validation objectives, not test-set optima: validation max- F_2 need not achieve the highest observed TEST F_2 .

Table 4

Threshold sensitivity on ULB 2013 — F_2 by threshold strategy.

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.618	0.771	0.771	0.708
RF	0.776	0.782	0.788	0.713
LGBM	0.765	0.771	0.813	0.730
CatBoost	0.784	0.784	0.775	0.740
FT-Trans.	0.766	0.762	0.761	0.780
OCSVM	0.375	0.481	0.485	0.433

Table 5Threshold sensitivity on BAF Base — F_2 by threshold strategy.

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.015	0.245	0.287	0.032
RF	0.001	0.263	0.294	0.045
LGBM	0.042	0.260	0.316	0.056
CatBoost	0.037	0.274	0.314	0.058
FT-Trans.	0.048	0.293	0.304	0.133
OCSVM	0.031	0.071	0.081	0.000

Table 6

Threshold sensitivity on BAF Base — Alert Rate (%) by threshold strategy.

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.02%	1.59%	4.14%	0.06%
RF	0.00%	1.55%	3.36%	0.08%
LGBM	0.07%	1.43%	3.25%	0.10%
CatBoost	0.05%	1.59%	3.80%	0.10%
FT-Trans.	0.07%	2.17%	5.71%	0.32%
OCSVM	1.03%	4.06%	7.32%	0.00%

On ULB, moving from fixed 0.5 to max- F_2 increases LR's F_2 from 0.618 to 0.771, approximately 25%, and LGBM's from 0.765 to 0.813. RF changes from 0.776 to 0.788, whereas CatBoost decreases from 0.784 to 0.775. FT-Transformer's max- F_2 rule gives 0.761, slightly below the fixed rule's 0.766. The highest supervised TEST F_2 changes from CatBoost under fixed 0.5 to LGBM under max- F_2 . These outcomes are compatible with validation-only selection: a DEV objective does not guarantee the largest TEST value. Each max- F_2 result exactly reproduces its baseline operating point.

On BAF, supervised fixed-threshold F_2 values range from approximately 0.001 to 0.048. The max- F_2 rules raise them to 0.287–0.316. LightGBM, CatBoost, and FT-Transformer improve by approximately 7.4, 8.5, and 6.3 times respectively. LR and RF have larger ratios because their fixed-rule starting values are particularly small; these ratios are descriptive within-model comparisons, not universal improvement bounds. For RF, the fixed rule flags one of 200,000 BAF cases, explaining both the small F_2 and the alert rate displayed as less than 0.01%.

The gain entails an operational cost. For supervised BAF baselines, max- F_2 produces alert rates of 3.25–5.71% and recall of 41.5–55.8%, rather than approximately 0.05–3.94% recall under fixed 0.5. For FT-Transformer specifically, recall rises from 3.94% to 55.80% while the alert rate increases from 0.0745% to 5.707%. This is 149 versus 11,414 alerts on the same 200,000 applications. A higher F_2 is therefore not a free improvement in investigation efficiency.

The validation precision-target rule gives lower BAF alert rates (approximately 0.06–0.32%) but also low recall (2.63–11.42%) for supervised models. Its TEST precision is not guaranteed to meet 0.5: LR reaches 0.487 and FT-Transformer 0.391 on BAF. The latter flags 645 applications, of which 252 are frauds. OCSVM has no feasible precision-target threshold in the BAF study and consequently issues no alerts. These outcomes support monitoring both achieved precision and absolute alert volume, rather than treating the rule's name as a deployment guarantee.

4.3. Imbalance Strategy by Model

Tables 7–10 retain the baseline-selected hyperparameters while varying the imbalance intervention. Figure 3 displays changes in ranking performance. Bold entries mark the largest unrounded point estimate in each row, including joint maxima when applicable; they do not denote statistically significant superiority.

On ULB, None produces the highest PR-AUC for LR, LGBM, CatBoost, and FT-Transformer. RF improves from 0.7928 to approximately 0.814 under SMOTE and SMOTE+Tomek. LGBM's rebalanced and weighted settings are closely grouped at approximately 0.811–0.814, below its 0.8205 baseline; RUS is substantially lower at 0.632. Diagnostics of all seven saved LGBM score vectors found no exact 0/1 probability saturation. Under ROS, SMOTE, SMOTE+Tomek, SMOTEENN and weighting, the five highest score groups contain only fraud cases, with small groups of one to eight rows, not mixed-label ties at the top. Thus the similar PR-AUC values and high ROC-AUC do

Table 7PR-AUC by model $\times$ strategy on ULB 2013. Bold: highest unrounded point estimate per row; SM+T: SMOTE+Tomek.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.692	0.619	0.669	0.672	0.672	0.683	0.670
RF	0.793	0.698	0.807	0.814	0.814	0.803	0.810
LGBM	0.820	0.632	0.811	0.814	0.814	0.811	0.813
CatBoost	0.820	0.637	0.811	0.798	0.798	0.793	0.802
FT-Trans.	0.801	0.641	0.784	0.785	0.785	0.797	0.756

Table 8 F_2 by model $\times$ strategy on ULB 2013 (max- F_2 threshold). Bold: highest unrounded point estimate per row; SM+T: SMOTE+Tomek.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.771	0.732	0.778	0.772	0.772	0.778	0.780
RF	0.788	0.774	0.812	0.805	0.805	0.753	0.821
LGBM	0.813	0.759	0.787	0.766	0.766	0.789	0.767
CatBoost	0.775	0.763	0.808	0.788	0.788	0.779	0.776
FT-Trans.	0.761	0.600	0.763	0.692	0.692	0.698	0.774

Table 9PR-AUC by model $\times$ strategy on BAF Base. Bold: highest unrounded point estimate per row; SM+T: SMOTE+Tomek.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.143	0.137	0.137	0.135	0.135	0.136	0.122
RF	0.159	0.135	0.149	0.120	0.120	0.124	0.146
LGBM	0.177	0.171	0.181	0.146	0.146	0.158	0.181
CatBoost	0.180	0.178	0.185	0.154	0.154	0.165	0.184
FT-Trans.	0.177	0.163	0.166	0.115	0.115	0.124	0.176

Table 10 F_2 by model $\times$ strategy on BAF Base (max- F_2 threshold). Bold: highest unrounded point estimate per row; SM+T: SMOTE+Tomek.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.287	0.283	0.283	0.280	0.280	0.282	0.267
RF	0.294	0.288	0.303	0.280	0.280	0.280	0.296
LGBM	0.316	0.318	0.319	0.293	0.293	0.300	0.320
CatBoost	0.314	0.317	0.316	0.303	0.303	0.309	0.317
FT-Trans.	0.304	0.301	0.310	0.257	0.257	0.269	0.313

not establish a common saturation mechanism or an implementation error. The two metrics describe different aspects of the recorded rankings. Class weighting reduces FT-Transformer PR-AUC from 0.8015 to approximately 0.756, and is not supported as a uniformly safe default.

On BAF, ROS gives CatBoost its highest PR-AUC, 0.185 versus 0.180 for None, an absolute increase of 0.005499 in the saved values. LightGBM's largest PR-AUC increase is 0.004048 under ROS. These modest ranking gains contrast with much larger losses from some synthetic interventions. SMOTE reduces FT-Transformer PR-AUC from 0.1769 to 0.1149 (approximately 35%) and CatBoost from 0.1796 to 0.1543 (approximately 14%). For the transformer, None has the highest BAF PR-AUC, whereas weighting has the highest F_2 (approximately 0.313), above ROS (0.310) and None (0.304). Ranking and decision criteria therefore favour different configurations.

The best ranking intervention need not give the highest F_2 . On BAF, LGBM with Class Weights yields the largest supervised F_2 in the BAF factorial (0.319575), whereas CatBoost with ROS has the largest PR-AUC. On ULB, RF with

Class Weights has the highest factorial F_2 (approximately 0.821), despite not having the highest PR-AUC. For CatBoost, RUS and Class Weights both round to $F_2 = 0.317$, but RUS has the larger unrounded value. The SMOTE/Tomek audit records zero Tomek removals and identical resampled inputs and TEST score arrays in every new ULB supervised comparison and the BAF FT-Transformer comparison. For the classical ULB models this covers all five validation folds and the full-DEV fit. For example, LGBM retains 453,204 full-DEV resampled rows and BAF FT-Transformer retains 1,582,354 rows under either sampler. Preserved BAF classical comparisons have identical reported metrics; archived full-DEV reconstruction also finds no removals, but does not supply original fold-level diagnostics. This explains the observed equality within its audited scope, not a general inability of Tomek Links to remove observations.

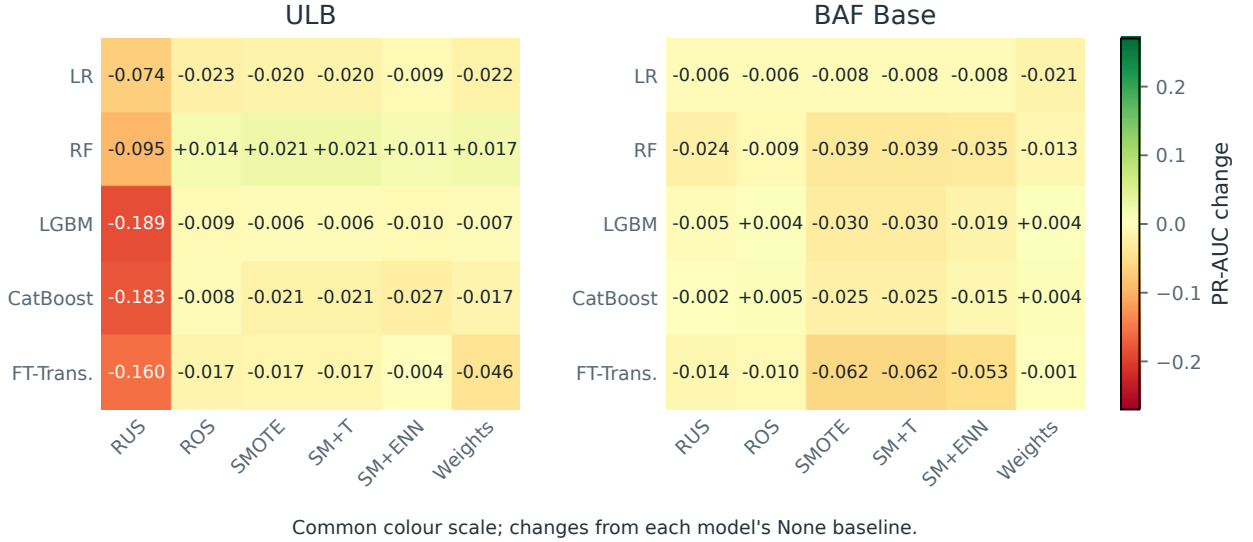


Figure 3: PR-AUC change relative to None on ULB (left) and BAF Base (right). Both panels use the same colour scale. Green indicates improvement and red indicates degradation; SM+ENN denotes SMOTEENN. These are descriptive point-estimate differences for the recorded runs, not significance tests. Plotting code was prepared with OpenAI Codex assistance and uses the saved experimental records.

4.4. Predefined Representation Controls

Table 11 reports supplementary BAF controls separately from the 72 primary configurations. Converting the documented absence codes to NaN and adding indicators increases LR PR-AUC from 0.1432 to 0.1477; its paired interval for the difference excludes zero. The remaining supervised PR-AUC intervals include zero, and OCSVM decreases from 0.0190 to 0.0162. The F_2 differences for all five supervised models have paired intervals including zero. This joint conversion/imputation/indicator intervention does not isolate a causal missingness effect or establish equivalence.

For the categorical-sampling control, SMOTE-NC yields FT-Transformer PR-AUC 0.1157 versus 0.1149 for primary SMOTE, with a paired interval including zero. It does not restore the None baseline's 0.1769. CatBoost decreases from 0.1543 to 0.1433. Its classifier still receives one-hot encoded inputs. These controls bound explanations of the primary SMOTE loss; neither is a matched architecture ablation, and no primary policy is selected using TEST.

The paired intervals describe TEST-row uncertainty conditional on the fitted pair, fixed DEV thresholds and split; they do not include training-seed variability. Historical CatBoost SMOTE pairing uses the verified raw-file identity, documented seed-42 split and exactly matched ordered labels; an unrecorded historical TEST index array is not assumed. Its archived six-decimal threshold is retained and must reproduce the saved confusion counts.

4.5. Uncertainty and Computational Cost

Table 12 reports BAF bootstrap intervals. CatBoost's PR-AUC interval is [0.164, 0.197] and FT-Transformer's is [0.162, 0.195]. The broad overlap warrants caution about small point-estimate differences; it does not prove equivalence or replace a paired comparison. BAF TEST contains 2,206 frauds, not the 11,029 frauds in the full dataset. ULB TEST contains only 95 frauds, further limiting the precision of comparisons on that benchmark.

Table 11

Predefined BAF representation controls, separate from the primary factorial. PR-AUC differences are alternative minus primary; intervals use 1,000 row-paired bootstrap resamples on the same TEST observations with fixed hyperparameters and separately DEV-selected thresholds.

Model	Primary	Alternative	Δ PR-AUC [paired 95% CI]
<i>Preserved absence codes $\rightarrow$ NaN + indicators</i>			
LR	0.1432	0.1477	+0.0045 [+0.0017, +0.0074]
RF	0.1588	0.1596	+0.0008 [−0.0015, +0.0030]
LGBM	0.1768	0.1759	−0.0009 [−0.0029, +0.0011]
CatBoost	0.1796	0.1790	−0.0005 [−0.0023, +0.0015]
FT-Trans.	0.1769	0.1732	−0.0038 [−0.0082, +0.0006]
OCSVM	0.0190	0.0162	−0.0028 [−0.0035, −0.0024]
<i>Primary SMOTE $\rightarrow$ SMOTE-NC</i>			
CatBoost	0.1543	0.1433	−0.0110 [−0.0147, −0.0074]
FT-Trans.	0.1149	0.1157	+0.0008 [−0.0068, +0.0076]

Table 12

95% Bootstrap CI for PR-AUC and ROC-AUC on BAF Base (strategy = None, $B=1000$).

Model	PR-AUC	95% CI	ROC-AUC	95% CI
LR	0.143	[0.131, 0.158]	0.875	[0.867, 0.882]
RF	0.159	[0.145, 0.175]	0.878	[0.870, 0.885]
LGBM	0.177	[0.162, 0.194]	0.898	[0.892, 0.905]
CatBoost	0.180	[0.164, 0.197]	0.898	[0.892, 0.904]
FT-Trans.	0.177	[0.162, 0.195]	0.896	[0.890, 0.903]
OCSVM	0.019	[0.018, 0.021]	0.628	[0.616, 0.639]

Table 13

Recorded BAF Base baseline workflow costs. † marks historical tuning/final-fit measurements retained with a frozen model, not a new fit or search. A dash denotes no new search for fixed or untuned models. Validation recovery costs are recorded separately and excluded from final-fit time.

Model	Tuning (s)	Train (s)	Inference (s)
LR	55,070 [†]	83.9 [†]	0.209
RF	105,313 [†]	1494.0 [†]	10.642
LGBM	1,243 [†]	16.1 [†]	1.298
CatBoost	7,010 [†]	36.7 [†]	0.213
FT-Trans.	—	354.4	4.262
OCSVM	—	919.6 [†]	71.579 [†]

Table 13 describes the recorded workflows, not a matched architecture benchmark. LGBM’s historical final fit took 16.1 seconds and CatBoost’s 36.7 seconds; their retained models attain PR-AUC 0.1768 and 0.1796. The revised FT-Transformer full-DEV fit took 354.4 seconds and attains 0.1769. Its retained hyperparameters require no new search, so the dash is not a claim that neural tuning is free. Historical classical searches and current transformer fitting do not share a measurement period or validation workload. Current full-TEST inference times are 1.298 seconds for LGBM, 0.213 seconds for CatBoost and 4.262 seconds for FT-Transformer. These measurements can inform resource planning only with their recorded hardware, implementation and provenance; they do not establish intrinsic speed or total investigation cost.

5. Discussion

5.1. The Contribution: Separate Ranking from Decisions

The main finding is a within-model operating-point effect. In the threshold study, each model’s PR-AUC is constant across rules because its saved TEST ranking is unchanged, yet F_2 , recall, and alert volume can change substantially. This directly supports treating threshold selection as part of detector design. It does not establish that thresholds matter more than classifiers in every task, nor require comparing the numerical range of PR-AUC with that of F_2 , which are different metrics answering different questions.

The low fixed-threshold recall on BAF is consistent with many fraud cases receiving scores below 0.5. It does not show that all fraud scores occupy a narrow interval, or that imbalance alone causes this behaviour. Figure 1 provides empirical transformer score distributions, while the operating-point tables establish the broader supervised-model result. The validation-selected BAF max- F_2 thresholds for supervised models are approximately 0.053–0.064; these values are decision cuts, not the limits of the fraud-score distribution.

Moving the cut changes which cases enter the alert queue, not how the model ranks them. The resulting recall increase is useful only if the organisation can support the associated workload. Conversely, a high-precision rule may leave too much fraud unflagged. A production decision should therefore start from an explicit cost or capacity objective and validate the corresponding rule, rather than adopting either 0.5 or max- F_2 as an automatic universal prescription.

5.2. Interpreting Imbalance Effects Conservatively

The factorial supports a strong unmodified baseline and targeted validation of interventions, not blanket endorsement of class weighting or resampling. Ranking and decision performance respond differently: a strategy can improve F_2 at its selected cut while degrading PR-AUC. Because hyperparameters were selected under None, these results measure the effect of the implemented intervention at a fixed selected configuration; independently retuning each combination might change the outcome.

The larger BAF SMOTE loss for FT-Transformer than for CatBoost is an important pipeline-level observation. It is not a causal demonstration that attention is inherently more vulnerable to synthetic data. Architecture, optimisation, and categorical representation differ between the two paths. Interpolating ordinal category codes and converting them back to integers is a specific limitation of the transformer resampling pipeline. Categorical-aware alternatives and controlled ablations would be needed to separate these explanations. The predefined SMOTE-NC control in Section 4.4 does not recover the transformer’s baseline ranking, so nominal-aware sampling alone is not an established explanation of the loss. Neither improved nor degraded performance in this study establishes a general rule for all tabular transformers.

The saved sampler diagnostics explain why the additional Tomek stage gives the same result in the new runs: it removes no rows after SMOTE in those realised training partitions. This is conditional on the data, representation, random seed and sampler settings. It is neither evidence that the two algorithms are interchangeable nor a claim that extreme imbalance prevents Tomek pairs.

5.3. Evidence-Bounded Configuration Guidance

Table 14 provides starting points for validation, not deployment guarantees. Rows are tied to the evaluated dataset and objective; dataset size alone is not used as a substitute for ULB or BAF. In particular, no resampled configuration is combined with a threshold rule that was tested only under None.

Table 14

Candidate configurations supported by the recorded experiments. Each row names the evidence scope and objective; transfer to a new application requires fresh validation.

Evidence scope and objective	Model	Strategy	Threshold
ULB: highest factorial F_2	RF	Weights	max- F_2
ULB: highest baseline-rule F_1	LGBM	None	max- F_2
BAF: highest factorial F_2	LGBM	Weights	max- F_2
BAF: highest baseline-rule F_1	CatBoost	None	max- F_1
BAF: precision-target recall candidate	FT-Trans.	None	Prec. target
BAF: short historical final-fit candidate	LGBM	None	max- F_2
BAF: highest transformer PR-AUC	FT-Trans.	None	max- F_2

For an application resembling BAF and prioritising recall-weighted performance, LGBM with weighting is an evidence-based candidate, but the small gain over other max- F_2 baselines should not be treated as a proven performance margin. If the investigator objective instead favours F_1 , CatBoost with None is supported by the baseline threshold study; CatBoost with ROS and max- F_1 was not tested and is not recommended as an established result. For a validation precision target, FT-Transformer has the highest supervised BAF recall among the tested target-rule configurations, but only 11.42% recall and approximately 0.32% alert rate; its achieved TEST precision is 0.391. This is a validation-constrained operating point, not a broadly high-recall solution.

The cost-oriented row concerns a historical final fit in the recorded setting, not inference throughput or total business cost. CatBoost’s recorded BAF inference time is lower than LGBM’s, illustrating why an unqualified speed ranking would be too broad a conclusion. The transformer row retains None for its strongest evaluated BAF ranking, not for the largest F_2 . Weighting has a higher transformer F_2 in the factorial and should be assessed against its alert workload rather than preferred simply because it avoids synthetic interpolation.

5.4. Limitations and Scope of Generalisation

The study uses two public benchmarks and one fixed outer split per dataset. ULB covers a short transaction period, and BAF is synthetic. Random stratification does not simulate prospective temporal deployment, feedback delays, or adversarial adaptation. No claim of production robustness, fairness, or causal architectural superiority follows from these experiments.

Selection variability is also incompletely characterised. The 50-trial budget covers different search spaces, the transformer uses one internal holdout rather than five folds, and the factorial reuses baseline hyperparameters. ULB is searched afresh after profile deduplication, whereas BAF retains selected hyperparameters and classical final fits. CatBoost does not use native categorical inputs. Test bootstraps do not address these sources of variation. The four threshold rules deliberately share each baseline’s saved validation scores and TEST ranking, isolating the decision rule rather than refitting effects.

Finally, F_2 is a convenient recall-weighted criterion rather than a monetary utility function. Alert rate is a workload proxy, not a measurement of analyst time, customer harm, or prevented losses. Recommendations in Table 14 are post-hoc summaries of benchmark evidence, not independently validated configuration-selection rules. Repeated-seed, temporal, and capacity-aware evaluations would strengthen external validity, but were not performed and are not implied by the present results.

6. Conclusion

This study separates model ranking, imbalance intervention, and threshold selection in a comparison of classical models and FT-Transformer on ULB and BAF Base. Its central empirical contribution is to show the operational consequence of applying explicit validation-selected rules to unchanged TEST scores, while retaining the alert workload needed to interpret the gain. Model and threshold selection remain within DEV, outside the held-out TEST sets.

On BAF, supervised max- F_2 rules yield $F_2 = 0.287\text{--}0.316$ compared with 0.001–0.048 under fixed 0.5. For FT-Transformer, the approximately 6.3-fold increase accompanies an alert-rate rise from 0.07% to 5.71%. The imbalance study offers no universal remedy: unmodified baselines are often strong, and synthetic resampling can reduce ranking quality substantially. Boosting remains a competitive reference with short recorded classical final-fit times, but cost comparisons retain their workflow qualifications. The results neither establish neural superiority nor exclude it under other representations, budgets, or datasets.

The practical implication is to validate the complete score-to-alert pipeline against an explicit operational objective. Future work should test temporal revalidation, capacity-constrained thresholds, repeated-seed comparisons, and matched categorical and preprocessing ablations. These extensions would address the current evidence limits without confusing a promising benchmark operating point with a deployment guarantee.

Data and Code Availability

The study uses the public ULB Credit Card Fraud Detection dataset (Machine Learning Group, Université Libre de Bruxelles, 2016) and BAF dataset suite (Feedzai Research, 2022). Code, reproduction instructions and compact aggregate evaluation evidence are available at <https://github.com/0lavo200100274/FraudDetection>. Dataset use remains subject to the respective providers’ terms. The authors retain the pinned per-row score/label arrays, row

identities and fitted models in an audit archive; these are not included in the compact public export. Additional access or deposit arrangements remain to be confirmed by the authors before journal submission. The results reported here reuse the recorded experimental runs; manuscript and figure revisions do not replace those results.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

OpenAI Codex was used to assist with manuscript revision, checks against the saved experimental evidence, and reproducible plotting and layout code for the figures. Experimental values were read from recorded evaluations, not invented or inferred by a generative model. The authors remain responsible for the accuracy, interpretation, and final approval of the manuscript.

CRedit authorship contribution statement

Olavo Caixeiro: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Maryam Abbasi:** Conceptualization, Supervision, Writing – review & editing. **Pedro Miguel de Oliveira Martins:** Supervision, Writing – review & editing.

References

- Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: A next-generation hyperparameter optimization framework, in: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, pp. 2623–2631. doi:<https://doi.org/10.1145/3292500.3330701>.
- Batista, G.E.A.P.A., Prati, R.C., Monard, M.C., 2004. A study of the behavior of several methods for balancing machine learning training data. ACM SIGKDD Explorations Newsletter 6, 20–29. doi:<https://doi.org/10.1145/1007730.1007735>.
- Breiman, L., 2001. Random forests. Machine Learning 45, 5–32. doi:<https://doi.org/10.1023/A:1010933404324>.
- Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P., 2002. SMOTE: Synthetic minority over-sampling technique. Journal of Artificial Intelligence Research 16, 321–357. doi:<https://doi.org/10.1613/jair.953>.
- Dal Pozzolo, A., Boracchi, G., Caelen, O., Alippi, C., Bontempi, G., 2018. Credit card fraud detection: A realistic modeling and a novel learning strategy. IEEE Transactions on Neural Networks and Learning Systems 29, 3784–3797. doi:<https://doi.org/10.1109/TNNLS.2017.2736643>.
- Dal Pozzolo, A., Caelen, O., Johnson, R.A., Bontempi, G., 2015. Calibrating probability with undersampling for unbalanced classification, in: 2015 IEEE Symposium Series on Computational Intelligence (SSCI), pp. 159–166. doi:<https://doi.org/10.1109/SSCI.2015.33>.
- Feedzai Research, 2022. Bank account fraud dataset suite. URL: <https://github.com/feedzai/bank-account-fraud>. dataset repository; accessed 22 July 2026.
- Fernández, A., García, S., Herrera, F., Chawla, N.V., 2018. SMOTE for learning from imbalanced data: progress and challenges, marking the 15-year anniversary. Journal of Artificial Intelligence Research 61, 863–905. doi:<https://doi.org/10.1613/jair.1.11192>.
- Gorishniy, Y., Rubachev, I., Khrulkov, V., Babenko, A., 2021. Revisiting deep learning models for tabular data, in: Advances in Neural Information Processing Systems (NeurIPS), pp. 18932–18943. URL: <https://papers.nips.cc/paper/2021/hash/9d86d83f925f2149e9edb0ac3b49229c-Abstract.html>.
- Hayat, K., Magnier, B., 2025. Data leakage and deceptive performance: A critical examination of credit card fraud detection methodologies. Mathematics 13, 2563. doi:<https://doi.org/10.3390/math13162563>.
- He, H., Garcia, E.A., 2009. Learning from imbalanced data. IEEE Transactions on Knowledge and Data Engineering 21, 1263–1284. doi:<https://doi.org/10.1109/TKDE.2008.239>.
- Jesus, S., Pombal, J., Alves, D., Cruz, A.F., Saleiro, P., Ribeiro, R.P., Gama, J., Bizarro, P., 2022a. BAF dataset suite datasheet. URL: <https://github.com/feedzai/bank-account-fraud/blob/main/documents/datasheet.pdf>. official documentation accompanying the NeurIPS 2022 Datasets and Benchmarks paper; accessed 7 October 2026.
- Jesus, S., Pombal, J., Alves, D., Cruz, A.F., Saleiro, P., Ribeiro, R.P., Gama, J., Bizarro, P., 2022b. Turning the tables: Biased, imbalanced, dynamic tabular datasets for ml evaluation, in: Advances in Neural Information Processing Systems, pp. 33563–33575. URL: <https://doi.org/10.52202/068431-2432>, doi:<https://doi.org/10.52202/068431-2432>. datasets and Benchmarks Track.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree, in: Advances in Neural Information Processing Systems (NeurIPS). URL: <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>.
- Luzio, E., Ponti, M.A., Arevalo, C.R., Argerich, L., 2024. Decoupling decision-making in fraud prevention through classifier calibration for business logic action, in: Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing (ACM-SAC), pp. 1036–1037. doi:<https://doi.org/10.1145/3605098.3636089>.
- Machine Learning Group, Université Libre de Bruxelles, 2016. Credit card fraud detection. URL: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>. dataset; observations collected in September 2013; first released on Kaggle in 2016; accessed 22 July 2026.
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., Gulin, A., 2018. CatBoost: unbiased boosting with categorical features, in: Advances in Neural Information Processing Systems (NeurIPS). URL: <https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>.

- Ryman-Tubb, N.F., Krause, P., Garn, W., 2018. How artificial intelligence and machine learning research impacts payment card fraud detection: A survey and industry benchmark. *Engineering Applications of Artificial Intelligence* 76, 130–157. doi:<https://doi.org/10.1016/j.engappai.2018.07.008>.
- Saito, T., Rehmsmeier, M., 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE* 10, e0118432. doi:<https://doi.org/10.1371/journal.pone.0118432>.
- Schölkopf, B., Platt, J.C., Shawe-Taylor, J., Smola, A.J., Williamson, R.C., 2001. Estimating the support of a high-dimensional distribution. *Neural Computation* 13, 1443–1471. doi:<https://doi.org/10.1162/089976601750264965>.
- Shwartz-Ziv, R., Armon, A., 2022. Tabular data: Deep learning is not all you need. *Information Fusion* 81, 84–90. doi:<https://doi.org/10.1016/j.inffus.2021.11.011>.